

OS, Embedded Basics, and Interview Tips

1. Operating Systems Basics ★ ★ ★ ★

Since she knows FreeRTOS and pthreads, interviewers may ask OS fundamentals.

Process vs Thread

Process	Thread
Independent program	Smallest execution unit
Own memory	Shares memory
Heavyweight	Lightweight

Context Switching

CPU switches from one task/thread to another.

Task A ↓

Task B ↓

Task C

Synchronization

Why do we need Mutex or Semaphore?

To avoid Race Conditions and Deadlocks.

She already knows this because of FreeRTOS.

2. ARM Cortex Basics ★ ★ ★

Her resume mentions Cortex-M3.

- ARM = RISC architecture
 - Low power
 - Widely used in embedded systems

Difference:

ARM Cortex-M → Microcontrollers (ESP32, STM32)

ARM Cortex-A → Application processors (Phones, Linux systems)

3. RISC vs CISC ★ ★ ★ ★

RISC	CISC
Simple instructions Fixed instruction size Faster execution Example: ARM	Complex instructions Variable instruction size Example: Intel x86

Interview Answer

"ARM processors use a RISC architecture with simpler instructions for better power efficiency, whereas x86 processors use CISC with more complex instructions."

4. Memory-Mapped I/O ★ ★ ★

Very common embedded question. CPU communicates with peripherals using memory addresses.

Example:

0x40021000 → GPIO Register

Writing "GPIO = HIGH" actually writes to a memory address.

5. Interrupts ★ ★ ★ ★

Instead of continuously checking sensors, hardware interrupts CPU.

Example:

Button pressed → Interrupt → CPU immediately executes ISR.

Know the difference between Polling vs Interrupt.

6. DMA (Surface Level)

DMA = Direct Memory Access

Normally: Sensor → CPU → RAM

DMA: Sensor → RAM (CPU doesn't participate.)

Benefits: Faster, Lower CPU usage.

7. Hardware Debugging

Know tools:

- Serial Monitor
- Logic Analyzer

- Oscilloscope

Even if she hasn't used an oscilloscope extensively, know what it is.

8. Git Basics

Since resume mentions Git. Know:

clone / commit / push / pull / branch / merge

9. Linux Basics ★ ★ ★ ★

Commands

ls / cd / pwd / grep / cat / chmod / ps / top / kill

Know Compilation Tools

gcc / g++ / make

Since Industrial project uses Makefile.

10. Basic OOP Revision

Even though C++ is done, revise:

- Abstraction
 - Encapsulation
 - Inheritance
 - Polymorphism

Interviewers love: "What is virtual function?"

11. Communication Skills ★ ★ ★ ★ ★

She should answer using structure.

Instead of:	Say:
"I used FreeRTOS."	"I built a four-task FreeRTOS application where different tasks handled sensing, display, and networking. Queues were used for inter-task communication and mutexes protected shared serial output."

Questions From Resume

These are more likely than SystemC. Expect:

- Explain your Healthcare project.
- Why Queue? Why Mutex?
- Why Dual Core? Why ESP32?
- How does MAX30102 work?
- Explain pthread project.
- Explain race condition.
- Explain one bug you solved.

Behavioral Questions

Prepare: Tell me about yourself. (2 minutes)

Prepare: Tell me about your internship.

Why Infineon?

Good answer:

"Infineon is a leader in automotive, power electronics, and embedded semiconductor solutions. I'm interested in low-level software and embedded systems, and this role gives me an opportunity to work on SystemC-based virtual prototyping and next-generation SoC development while learning from experienced engineers."

Why should we hire you?

Good answer:

"I have a strong embedded systems foundation through projects on ESP32, ARM Cortex-M3, Linux, and FreeRTOS. While technologies like SystemC are new to me, I'm comfortable learning new tools quickly, and I believe my problem-solving skills and firmware experience will help me contribute effectively."

Tell me a difficult bug.

She has several:

- Race condition in FreeRTOS
Shared serial buffer
Queue synchronization

Those make excellent stories.

If They Ask Something Unknown

The best answer:

"I haven't worked with that technology directly, but based on my understanding..." followed by what she knows, and then "...I'd be excited to learn it in depth."

That demonstrates honesty and confidence.